Q1.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Figure 6 shows a device that removes small particles of soot from sooty air.

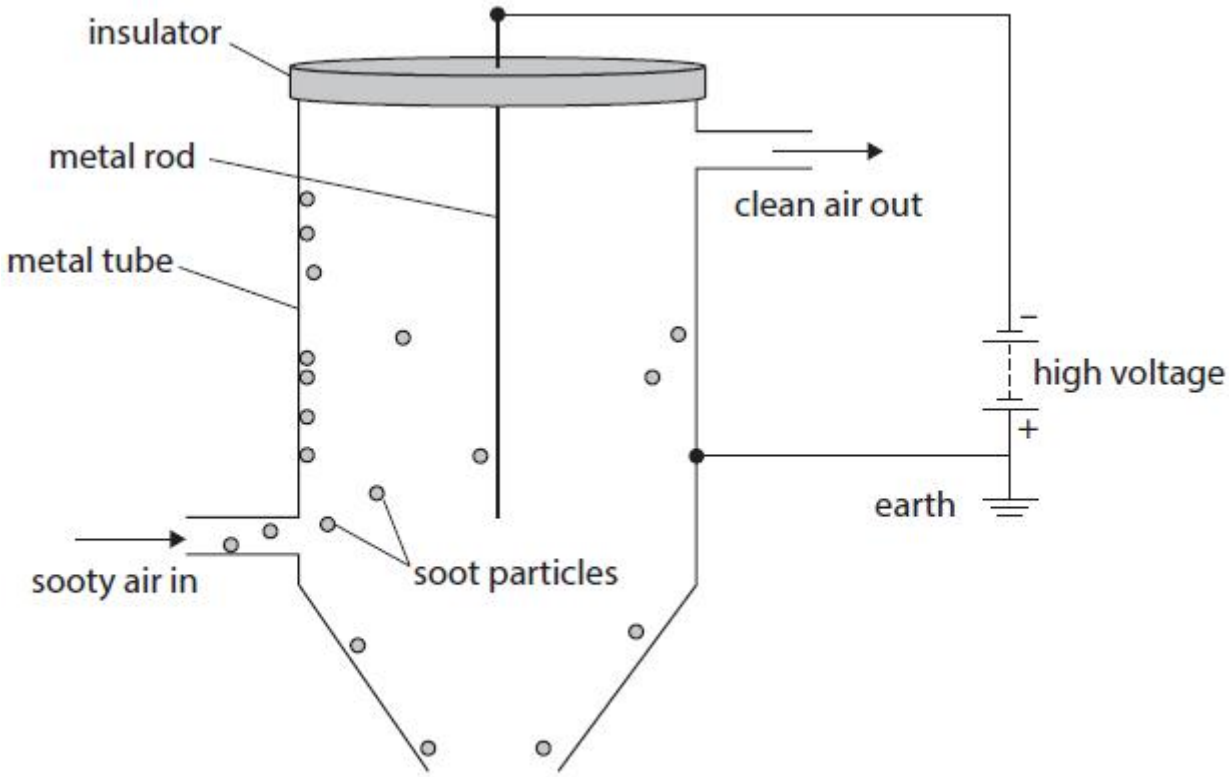

Figure 6

(i) There is a large potential difference between the metal tube and the metal rod in the centre of the tube.

This large potential difference causes the soot particles to gain a negative charge as they pass between the rod and the tube.
The soot particles have

(1)

☐ **A** lost protons
☐ **B** lost electrons
☐ **C** gained protons
☐ **D** gained electrons

(ii) Explain why, in Figure 6, the charged soot particles move towards the metal tube.

(2)

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.....

(iii) When the soot particles reach the metal tube, the charge on them changes.

State how the charge on them changes.

(1)

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(iv) The metal tube does not become charged when the soot particles reach it.

Explain why the metal tube does not become charged.

(2)

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(Total for question = 6 marks)

Examiner's Report

Q1.

(ii)
The charged soot particles are in an electric field and are attracted towards the metal tube. Alternatively a positive charge would be induced on the surface of the metal tube resulting in a force of attraction of the soot particles towards the metal tube. Examiners would accept an answer that linked attraction towards the metal tube and unlike charges or repulsion from the central metal rod because of like charges.

(ii) Explain why, in Figure 6, the charged soot particles move towards the metal tube.

(2)

- negative soot particles attract to ~~the~~ positive charged metal tubes.
- interacting electric field

Results Plus: Examiner Comments
This response scores a mark for attraction and another mark for reference to the positive metal tube.
There is no credit for reference to the negative charge on the soot particles because this had already been given in the previous part of the question.
It also correctly refers to an electric field. However, two out of two marks had been already been scored.

the positive charges in the metal tube are brought to the surface of the tube, attracting the negatively charged soot particles, as opposite charges attract.

Results Plus: Examiner Comments
A good description of positive charges on the surface of the metal rod attracting the oppositely charged soot particles. It scored two out of two marks.

The metal rod is negatively charged by the high voltage circuit. As the soot particles are also negatively charged, like charges repel and causes the soot to stick to the tube.

Results Plus: Examiner Comments
A clear answer that correctly described repulsion between the negative rod and soot particles. Two out of two scored.

they get repelled by the metal rod ~~beca~~ causing them to go to the tube.

Results Plus: Examiner Comments
This scores a mark for the idea of the central metal rod repelling the particles for one mark. To score two marks it should have stated that the rod had the same charge as the particles (ie negative).

(iii-iv)
In (iii), candidates were asked what happens to the charge on the soot particles when they reach the metal tube.
Examiners were looking for the idea of charge being lost from the soot particles as they settle on to the metal tube. This could be expressed as becoming neutral or being discharged. Most candidates were able to provide an acceptable response.
In (iv), the metal tube is a conductor that is connected to earth. Therefore, any electric charge transferred to it by the soot particles will travel through the metal and it's connected wire to earth. Very many candidates gave a good explanation along these lines.
A mark was sometimes lost by failing to state that the metal tube is a conductor.
Other candidates lost a mark by failing to recognise that charge would only move through the metal tube if it had somewhere to go, ie earth.

Mark Scheme

Q1.

Question number	Answer	Additional guidance	Mark
(i)	D gained electrons A and C are incorrect because protons are not exchanged in this context B is incorrect because it would give the particles a net positive charge		(1) AO1.1

Question number	Answer	Additional guidance	Mark
(ii)	an explanation linking any **two** from (soot particles are in an) electric field (1) **AND EITHER** attraction (1) (surface of) it/tube has (induced) positive/opposite/unlike charge (1) **OR** repulsion (1) rod (and soot particles) has a negative/like/same charge (1)		**(2)** **AO1.1**

Question number	Answer	Additional guidance	Mark
(iii)	(net charge) becomes zero (1)	discharged / charge is lost / charge moves on to the metal tube/ becomes neutral allow electrons for charge ignore becomes positively charged	**(1)** **AO1**

Question number	Answer	Additional guidance	Mark
(iv)	an explanation linking metal is a conductor (1) (connected to) earth/ground (1)	allow electrons for charge charge moves through the metal / tube / wire	**(2)** **AO1**